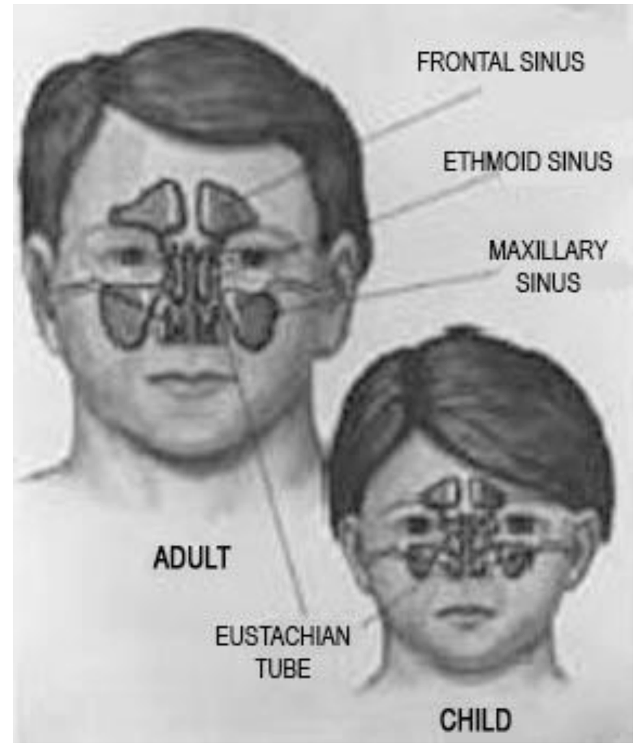
MORPHOLOGY OF THE PARANASAL SINUSES

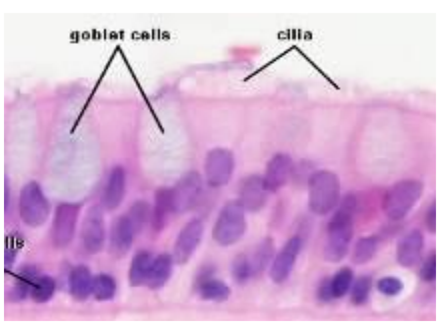
FUNCTION

- Warming / cooling of the inspired air
- „Buffering space”
- „Makes the head lighter” (pneumatized bones)
- Mucus production, moistening

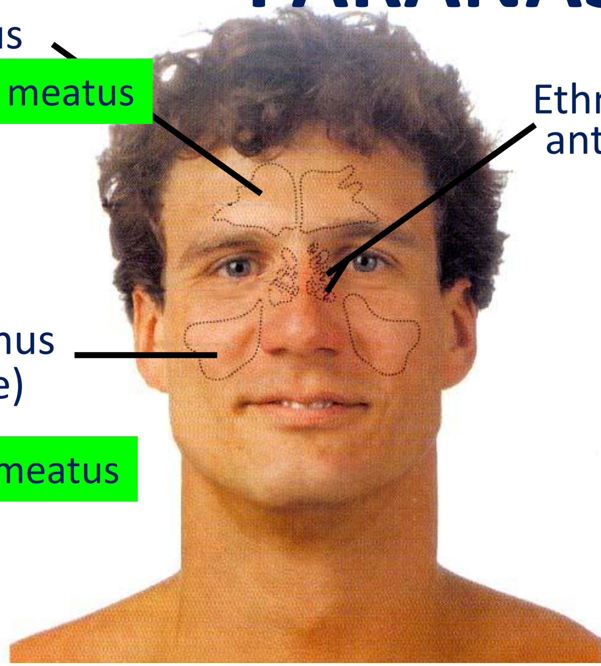


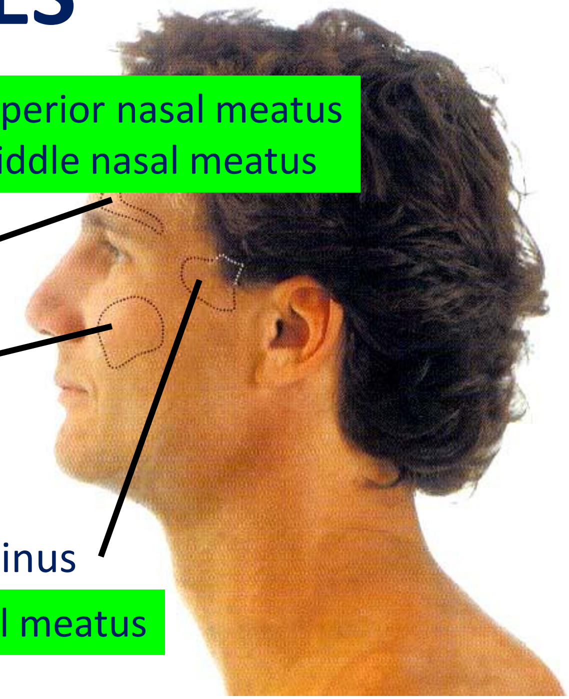
HISTOLOGY

Respiratory epithelium upon lamina propria
Goblet cells
Mixed merocrine glands (seromucous)

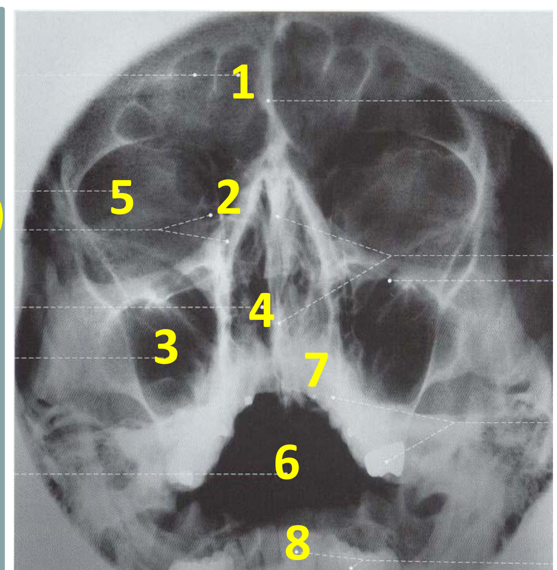


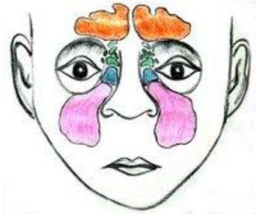
PARANASAL SINUSES





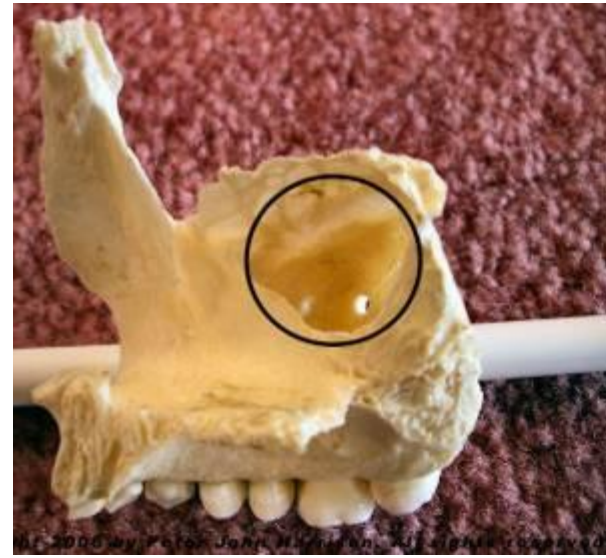
1. Frontal sinus
2. Ethmoidal air cells
3. Maxillary sinus (Highmore)
4. Common nasal meatus
5. Orbit
6. Oral cavity proper
7. Superior teeth
8. Inferior teeth

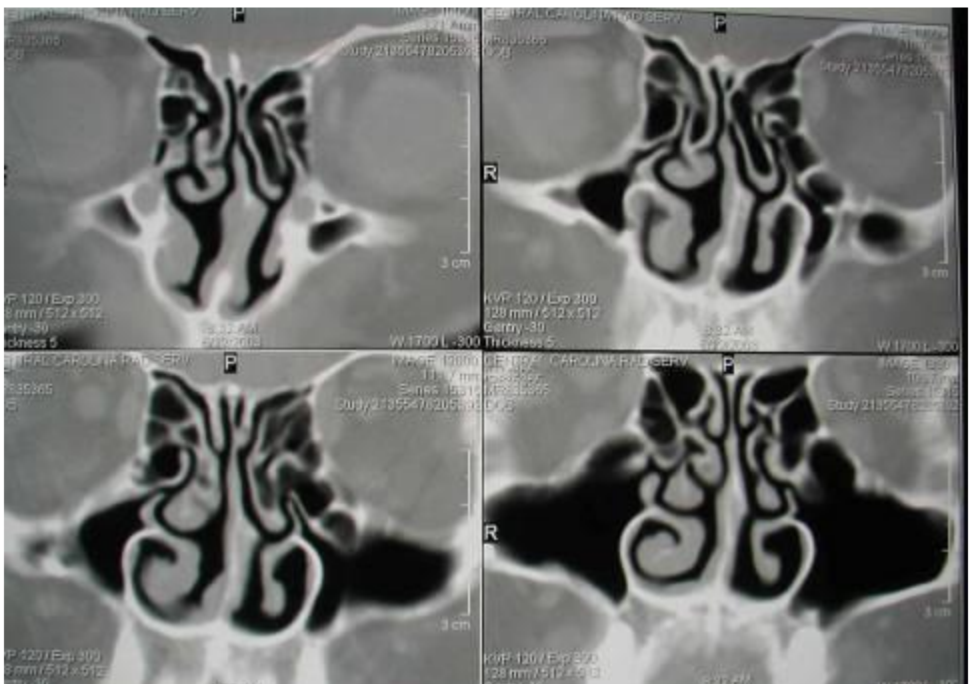


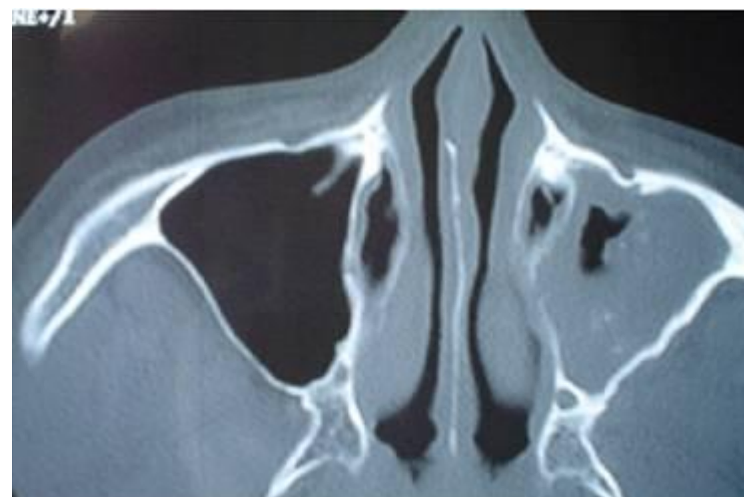


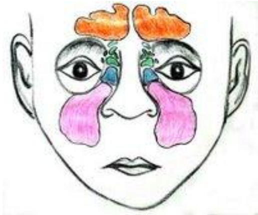
MAXILLARY SINUS

The largest sinus (of *Highmore*)
Opens via the semilunar hiatus
Innervation: maxillary nerve (*post, med sup alv, infraorbital branches)*
Important topographical relation:
Roots of the upper teeth and orbit

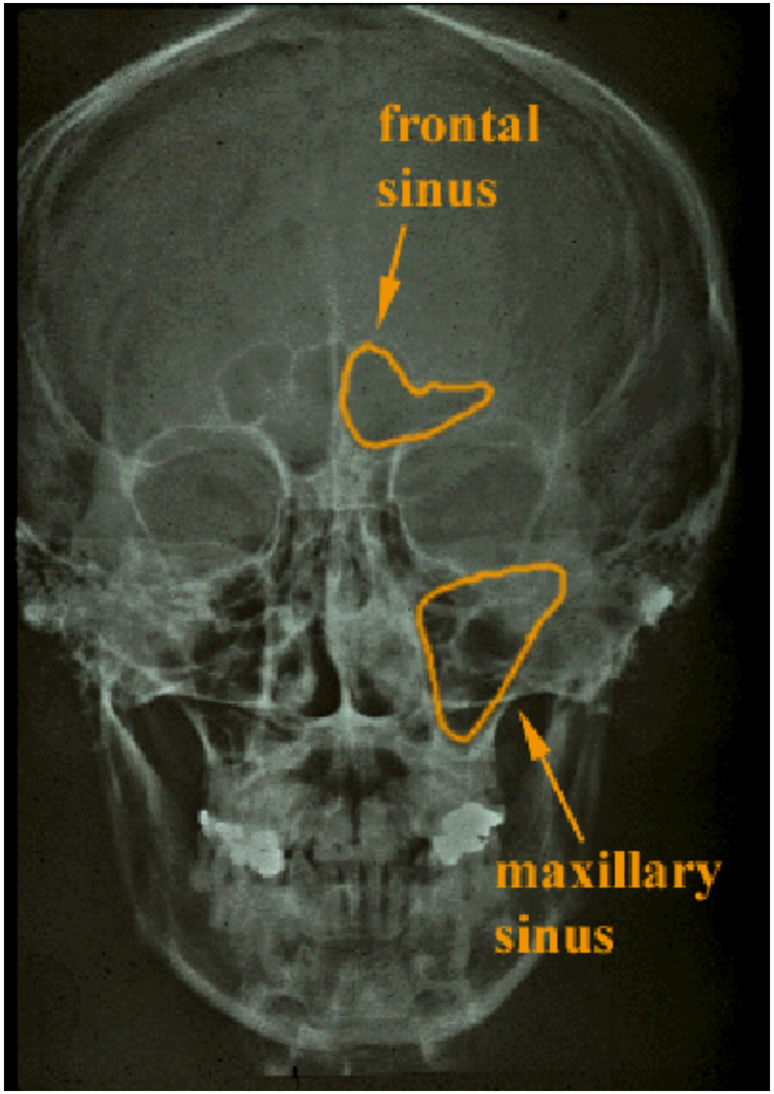






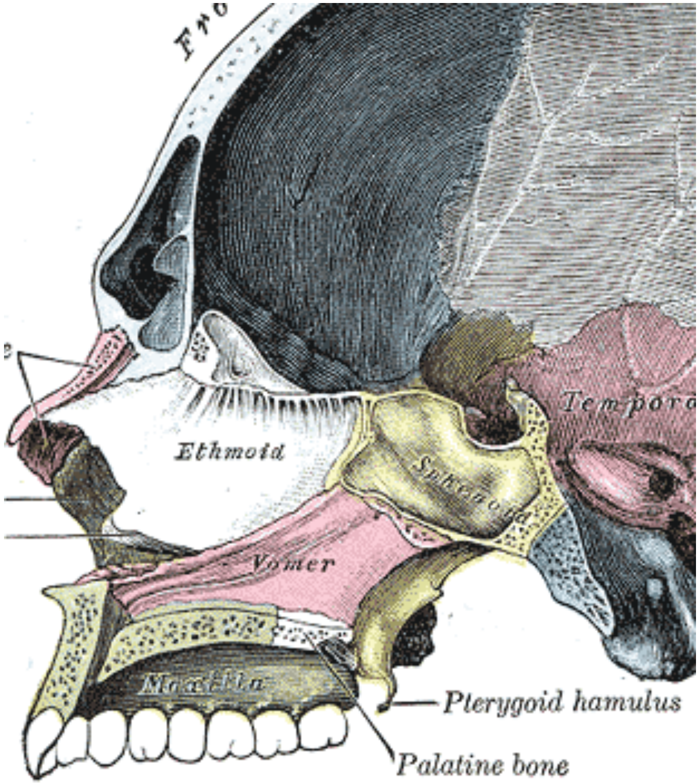


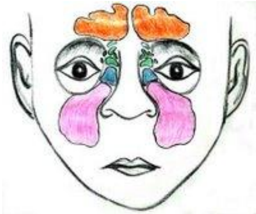
FRONTAL SINUS



Opens via the ethmoidal infundibulum (frontonasal duct) at the semilunar hiatus (anterior aspect)

Innervation: supraorbital & supratrochlear nerves (CN 5/1)



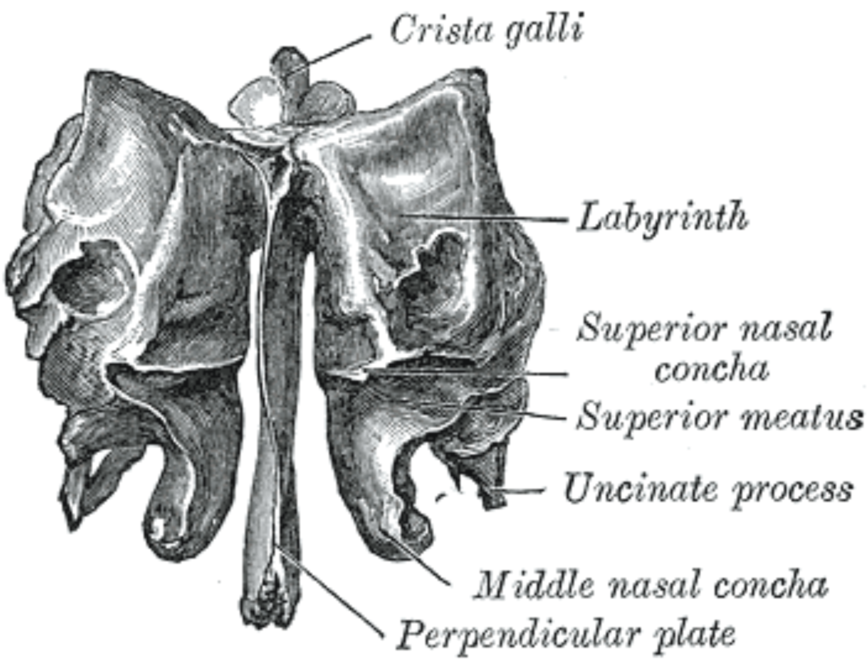


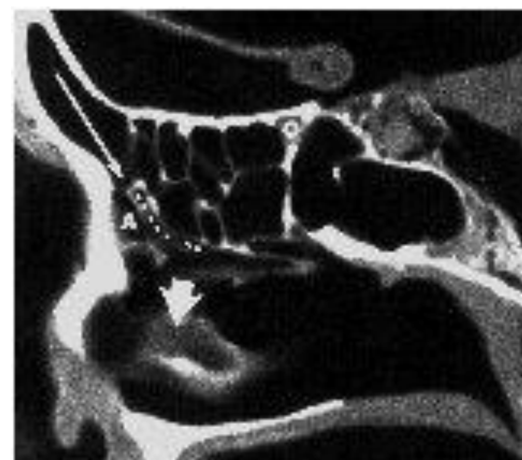
ETHMOIDAL SINUS (LABIRYNTH)

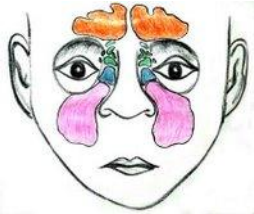
Numerous openings

- **anterior and medial air cells – semilunar hiatus**
- **posterior air cells - superior nasal meatus**

Innervation: ant & post ethmoidal nerves (CN5/1)







SPHENOIDAL SINUS

Paired cavities in the body of sphenoid
Open separately through the aperture of the sphenoidal sinus–
within the sphenoethmoidal recess
Innervation: sphenopalatine nerve (CN 5/2 + parasymp)